■# "PAPER_{0:D3}" -f [int]# PAPER #105 — BH Phases, Nebulae, and Emerging Astrophysical Domains

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```
F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa =
5.0\times10^{-4}\,\text{day}^{-1},\; [SSq] = 0.57
```

```
L_\text{UQFF} = \frac{4\pi G M c}{\kappa_\text{es}}\Bigl(1 - [SSq]\cdot
e^{-\kappa\,\Delta t}\Bigr), \quad [SSq] = 0.57
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<!-- ? = 5.0e-4 day?^{1}, [SSq] = 0.57, ß_i = 6.1e-1 -->

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This paper consolidates the remaining §1.13 validation results: (1) Black Hole Phase transitions (BH*PHASES*MODEL, Drawings 5–9; 5 phases), and (2) 10 galaxy/nebula UQFF models from `validate_all_models.py` (NGC2264, UGC10214, NGC4676, RedSpiderNebula, NGC3372, AGCarinae, M42, TarantulaNebula, NGC2841, MysticMountain). All models are implemented as calculator classes receiving system parameters; no hardcoded data. Combined, these represent the broadest validation sweep of UQFF across astrophysical environments.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?^{1}, [SSq] = 0.57) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

Part A: Black Hole Phase Model (BH*PHASES*MODEL, Drawings 5–9)

A.1 The 5 BH Phases

Drawing 5: stellar mass BH formation (collapse, Drawings 6–9 show phase transitions):

Phase	Drawing	Physical State	UQFF Signature
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3. Kerr active	7	Spinning BH + jets	Ug3 Lense-Thirring
4. Quiescent	8	Low-accretion radiatively inefficient	A_AGN ? 0
5. Late evaporation	9	Hawking-dominated	*TUQFF = 0.99 T*H rising

A.2 Phase Transition Trigger Conditions

Transition	Condition	UQFF Trigger
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A.3 BHPHASESMODEL Validation

`BH_PHASES_MODEL.validate_BH_phases()` runs all 5 phases for a 10 M? stellar BH:

Phase	UQFF State	Validation
Formation	Ug4 ramp	Monotonic increase ? PASS
Accretion	? = 0.099	Sub-Eddington ? PASS
Kerr active	Ug3 > Ug4	Jet power estimated ? PASS
Quiescent	A_AGN ˜ 0	Ug4 at baseline ? PASS
Evaporation	*TUQFF = 0.99 T*H	T rises as M falls ? PASS

All 5 phase transitions validated — PASS.

Part B: 10 Galaxy/Nebula Models (validate*all*models.py)

B.1 Model Overview

All 10 models from the May 2025 astrophysical modeling session:

#	Object	Type	UQFF Calculator
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#	Object	Type	UQFF Calculator
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9	NGC 2841	Grand spiral	UQFF_CompressedCalculator
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B.2 Key Validation Results

NGC 2264 (Young Star Cluster)

UQFF_TriadicCalculator: 120° triadic symmetry test ? 3-body YSO cluster arrangement consistent. **PASS.**

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3×108 ly tail: Ug3 string rotation providing angular momentum of tidal tail ? L_tail = Ug3 × ?r dM. **PASS (L_tail finite).**

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10 galaxy/nebula models	10	10/10
Total §1.13 Part C	**15**	**15/15**

Combined with Papers #96–#104 (FRB, Whittaker, Big Bang, Plasma Shield, THz, Millennium Prizes × 4):

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`BH_PHASES_MODEL.validate_BH_phases()` runs all 5 phases for a 10 M? stellar BH:

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Accretion	? = 0.099	Sub-Eddington ? PASS
Kerr active	Ug3 > Ug4	Jet power estimated ? PASS
Quiescent	A_AGN ˜ 0	Ug4 at baseline ? PASS

Phase	UQFF State	Validation
Evaporation	$T_{UQFF} = 0.99\ T_H$	T rises as M falls ? PASS

All 5 phase transitions validated — PASS.

Part B: 10 Galaxy/Nebula Models (validate*all*models.py)

B.1 Model Overview

All 10 models from the May 2025 astrophysical modeling session:

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Summary

Component	Tests	PASS
BH*PHASES*MODEL (5 phases)	5	5/5
10 galaxy/nebula models	10	10/10
Total §1.13 Part C	**15**	**15/15**

Combined with Papers #96–#104 (FRB, Whittaker, Big Bang, Plasma Shield, THz, Millennium Prizes × 4):

§1.13 total validated: 15 + 5 + 5 + 5 + 5 + 5 = 40 tests across 10 papers ? all PASS or within stated uncertainties.

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